

1st order ODEs

Basic notions

differential equation : eqn. that involves a function $y(x)$ ($y(t)$) and its derivatives

implicit form :

$$F(x, y(x), y'(x), \dots, y^{(n)}(x)) = 0$$

↑
n'th order ODE

first order ODE :

$$(1) \quad F(x, y, y') = 0$$

implicit form

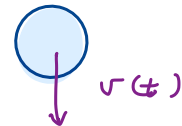
$$(2) \quad y' = f(x, y)$$

explicit form

e.g. dropping ball :

$$v'(t) = g - k \cdot v^2$$

↑
gravitation ↑
friction



$h(x)$ is a solution if it obeys (1) or (2)
on an open interval $]a, b[$

example:

$$(3) \quad y' = 0,4 \cdot y$$

a solution is $h(x) = e^{0,4 \cdot x}$;

$$h'(x) = 0,4 \cdot e^{0,4 \cdot x} = 0,4 \cdot h$$

ODE's have many solutions

$e^{0,4 \cdot x}$ is a particular solution of (3)

its general solution is $h(x) = c \cdot e^{0,4 \cdot x}$
 free parameter

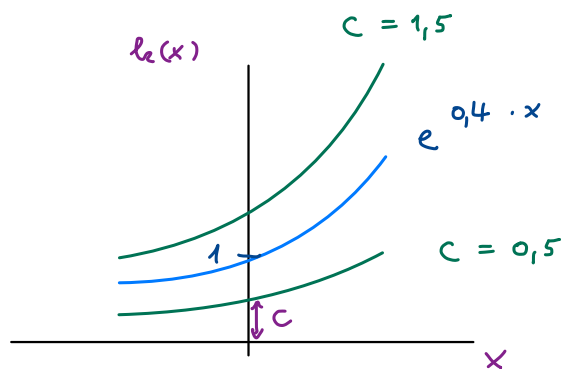


diagram of solutions

initial value problem:

find solution of (2) with $y(x_0) = y_0$

- initial value is satisfied by a particular solution, only

e.g. $y' = 0,4 \cdot y$; $y(1) = 3$

general solution : $c \cdot e^{0,4 \cdot x}$

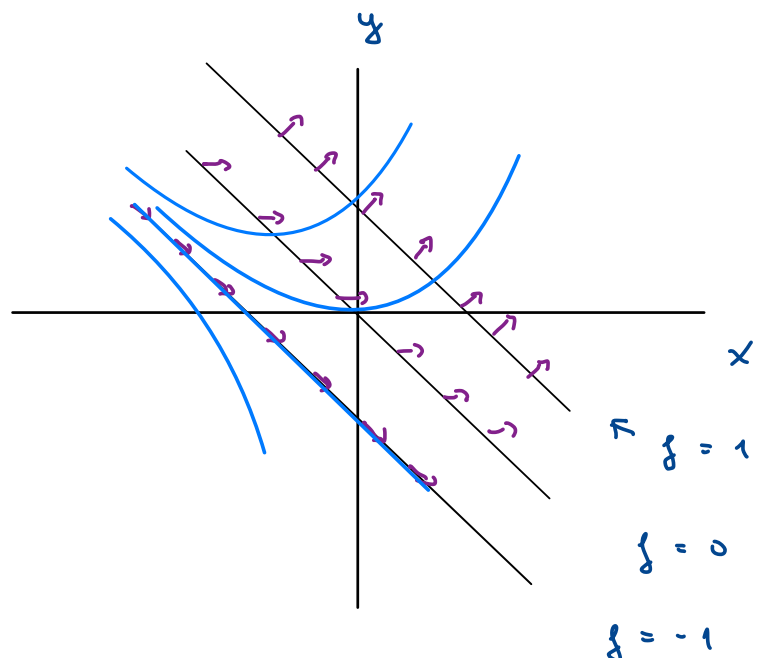
initial value : $c \cdot e^{0,4} = 3 \Rightarrow c = 3 \cdot e^{-0,4}$

$\Rightarrow h(x) = 3 \cdot e^{0,4 \cdot (x-1)}$ $\leftarrow$ particular solution satisfying $h(1) = 3$

direction fields

consider

$y' = y + x = f(x, y)$
 $\uparrow$ slope



$f = 1$

$f = 0$

$f = -1$

Numerical solution:

Euler's method:

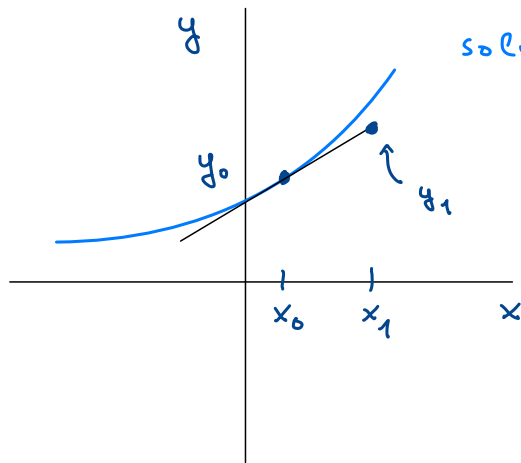
$$\Delta y = f(x, y) \cdot \Delta x \quad \Delta x \equiv h$$

$$x \Rightarrow x_0, x_0 + h, x_0 + 2h, \dots$$

$$y_0 \Rightarrow y_1 = y_0 + f(x_0, y_0) \cdot h$$

$$y_2 = y_1 + f(x_1, y_1) \cdot h$$

$\vdots$



becomes more and more precise as $h \rightarrow 0$!

Separable equations

can be written as

$$g(y) \cdot y' = f(x)$$

solution:

$$\int dx \, g(y) y' = \int dx \, f(x) + C$$

$$dx \cdot y' = dy$$

$$\int dy \, g(y) = \int dx \, f(x) + C$$

examples:

$$1. \quad y' = 0,4 \cdot y \Rightarrow \frac{1}{y} y' = 0,4 \Rightarrow$$

$$\int \frac{dy}{y} = \int 0,4 \cdot dx + \tilde{C} \Rightarrow \ln y = 0,4 \cdot x + \tilde{C}$$

$$\Rightarrow y(x) = C \cdot e^{0,4 \cdot x}$$

$$2. \quad v' = 10 - 0,1 \cdot v^2 \quad [v] = \frac{\text{m}}{\text{s}} \quad [g] = \left[\frac{\text{m}}{\text{s}^2} \right]$$

$$\int \frac{dv}{10 - 0,1 v^2} = \int dt + C$$

$$t + \tilde{C} = -\frac{1}{0,1} \int \frac{dv}{v^2 - 100} \stackrel{v > 0}{=} -\frac{1}{2} \ln \frac{v-10}{v+10} \Rightarrow$$

$$\frac{v-10}{v+10} = C e^{-2 \cdot t} \Rightarrow v = 10 \cdot \frac{1 + C \cdot e^{-2t}}{1 - C \cdot e^{-2t}} \quad \text{check!}$$

$$\text{initial condition: } v_0 = 10 \quad \frac{1+C}{1-C} \Rightarrow C = \frac{v_0 - 10}{v_0 + 10}$$

$$v < 10 \quad \dots \quad v = 10 \cdot \frac{1 - C e^{-2t}}{1 + C e^{-2t}}$$

